

# SQL-Lab8

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Consider the schema given below and answer the following queries in SQL

```
create table employee (fname varchar(50),minit char(1),lname varchar(50),
    ssn int primary key,bdate date,address varchar(100),sex char(1),
    salary decimal(10,2),superssn int,depno int);

create table department (dname varchar(50),depno int primary key,
    mgrssn int,mgrsdate date,foreign key (mgrssn) references employee(ssn));

alter table employee
add constraint fk_employee_super
foreign key (superssn) references employee(ssn);

alter table employee
add constraint fk_employee_department
foreign key (depno) references department(depno);

create table dept_locations (depno int,dlocation varchar(50),primary key (depno, dlocation),
    foreign key (depno) references department(depno));

create table project (pname varchar(50),pnumber int primary key,plocation varchar(50),depno int,
    foreign key (depno) references department(depno));

create table works_on (essn int,pno int,hours decimal(4,2),primary key (essn, pno),
    foreign key (essn) references employee(ssn),
    foreign key (pno) references project(pnumber));

create table dependent (essn int,dependent_name varchar(50),sex char(1),bdate date,relationship varchar(50),
    primary key (essn, dependent_name),
    foreign key (essn) references employee(ssn));
```

1. Display employee names ordered by department number and then by last name.

```
80 select fname, lname, depno from employee order by depno, lname;
81
```

| Data Output |                                 |                                 |                  | Messages | Notifications |
|-------------|---------------------------------|---------------------------------|------------------|----------|---------------|
|             |                                 |                                 |                  | Showing  |               |
|             | fname<br>character varying (50) | lname<br>character varying (50) | depno<br>integer |          |               |
| 1           | Mary                            | Johnson                         | 1                |          |               |
| 2           | John                            | Smith                           | 1                |          |               |
| 3           | Patricia                        | Brown                           | 2                |          |               |
| 4           | James                           | Williams                        | 2                |          |               |
| 5           | Linda                           | Davis                           | 3                |          |               |
| 6           | Robert                          | Jones                           | 3                |          |               |
| 7           | Michael                         | Miller                          | 3                |          |               |
| 8           | William                         | Moore                           | 4                |          |               |
| 9           | Barbara                         | Wilson                          | 4                |          |               |
| 10          | David                           | Anderson                        | 5                |          |               |
| 11          | Elizabeth                       | Taylor                          | 5                |          |               |

2. Find total number of hours worked on each project for each employee.

```

82 select E.fname, E.lname, P.pname, sum(w.hours) as total_hours from employee E
83 join works_on W on E.ssn = w.essn
84 join project P on W.pno = P.pnumber
85 group by E.fname, E.lname, P.pname;
86

```

|   | fname<br>character varying (50) | lname<br>character varying (50) | pname<br>character varying (50) | total_hours<br>numeric |
|---|---------------------------------|---------------------------------|---------------------------------|------------------------|
| 1 | David                           | Anderson                        | billing system                  | 7.00                   |
| 2 | John                            | Smith                           | ai system                       | 10.00                  |
| 3 | Barbara                         | Wilson                          | recruitment portal              | 20.00                  |
| 4 | William                         | Moore                           | recruitment portal              | 8.00                   |
| 5 | Mary                            | Johnson                         | ai system                       | 5.00                   |
| 6 | Linda                           | Davis                           | robotics                        | 6.00                   |
| 7 | Robert                          | Jones                           | robotics                        | 15.00                  |
| 8 | Elizabeth                       | Taylor                          | billing system                  | 18.00                  |
| 9 | James                           | Williams                        | payroll app                     | 12.00                  |

3. Display each department along with the average employee salary.

```

87 select D.dname, avg(E.salary) as avg_salary
88 from department D
89 join employee E on D.depno = E.depno
90 group by D.dname;
91

```

|   | dname<br>character varying (50) | avg_salary<br>numeric |
|---|---------------------------------|-----------------------|
| 1 | Research                        | 57500.000000000000    |
| 2 | Administration                  | 49000.000000000000    |
| 3 | Finance                         | 57000.000000000000    |
| 4 | Hr                              | 49000.000000000000    |
| 5 | Technical                       | 55666.666666666667    |

4. Display departments having more than 5 employees.

```

92 select d.dname, count(e.ssn) as emp_count from department d
93 join employee e on d.depno = e.depno group by d.dname
94 having count(e.ssn) > 5;
95

```

|  | dname<br>character varying (50) | emp_count<br>bigint |
|--|---------------------------------|---------------------|
|--|---------------------------------|---------------------|

5. Show departments whose average salary exceeds 50,000.

```

197 select d.dname, avg(e.salary) as avg_salary from department d join employee e on d.deptno = e.deptno
198 group by d.dname having avg(e.salary) > 50000;
199

```

|   | dname<br>character varying (50) | avg_salary<br>numeric |
|---|---------------------------------|-----------------------|
| 1 | Research                        | 57500.000000000000    |
| 2 | Finance                         | 57000.000000000000    |
| 3 | Technical                       | 55666.666666666667    |

6. Show employee name and project name for each employee who works on a project.

```

100 select e.fname, e.lname, p.pname from employee e join works_on w on e.ssn = w.essn
101 join project p on w.pno = p.pnumber;
102

```

|   | fname<br>character varying (50) | lname<br>character varying (50) | pname<br>character varying (50) |
|---|---------------------------------|---------------------------------|---------------------------------|
| 1 | John                            | Smith                           | ai system                       |
| 2 | James                           | Williams                        | payroll app                     |
| 3 | Robert                          | Jones                           | robotics                        |
| 4 | Barbara                         | Wilson                          | recruitment portal              |
| 5 | Elizabeth                       | Taylor                          | billing system                  |
| 6 | Mary                            | Johnson                         | ai system                       |
| 7 | Linda                           | Davis                           | robotics                        |
| 8 | William                         | Moore                           | recruitment portal              |
| 9 | David                           | Anderson                        | billing system                  |

7. List employees who earn more than their supervisors.

```

103 select e.fname, e.lname from employee e join employee s on e.superssn = s.ssn
104 where e.salary > s.salary;
105

```

|   | fname<br>character varying (50) | lname<br>character varying (50) |
|---|---------------------------------|---------------------------------|
| 1 | William                         | Moore                           |

8. Find all pairs of employees who work in the same department.

```

106 select e1.fname as emp1_fname, e1.lname as emp1_lname,
107 e2.fname as emp2_fname, e2.lname as emp2_lname, e1.deptno
108 from employee e1 join employee e2 on e1.deptno = e2.deptno and e1.ssn < e2.ssn order by e1.deptno;
109
110

```

|   | emp1_fname<br>character varying (50) | emp1_lname<br>character varying (50) | emp2_fname<br>character varying (50) | emp2_lname<br>character varying (50) | deptno<br>integer |
|---|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|-------------------|
| 1 | John                                 | Smith                                | Mary                                 | Johnson                              | 1                 |
| 2 | James                                | Williams                             | Patricia                             | Brown                                | 2                 |
| 3 | Robert                               | Jones                                | Michael                              | Miller                               | 3                 |
| 4 | Robert                               | Jones                                | Linda                                | Davis                                | 3                 |
| 5 | Linda                                | Davis                                | Michael                              | Miller                               | 3                 |
| 6 | Barbara                              | Wilson                               | William                              | Moore                                | 4                 |
| 7 | Elizabeth                            | Taylor                               | David                                | Anderson                             | 5                 |

9. Find employees who have at least one dependent.(use exists)

```

111 select e.fname, e.lname from employee e
112 where exists ( select 1 from dependent d where d.essn = e.ssn);
113

```

|   | fname<br>character varying (50) | lname<br>character varying (50) |
|---|---------------------------------|---------------------------------|
| 1 | John                            | Smith                           |
| 2 | James                           | Williams                        |
| 3 | Elizabeth                       | Taylor                          |

10. Find departments that do not control any project.

```

115 select d.dname from department d where not exists ( select 1 from project p
116 where p.depno = d.depno);
117

```

| dname<br>character varying (50) |
|---------------------------------|
|---------------------------------|

11. Find employees who have no dependents (use correlated NOT EXISTS).

```

119 select e.fname, e.lname from employee e where not exists ( select 1 from dependent d
120 where d.essn = e.ssn);
121
122

```

|   | fname<br>character varying (50) | lname<br>character varying (50) |
|---|---------------------------------|---------------------------------|
| 1 | Linda                           | Davis                           |
| 2 | William                         | Moore                           |
| 3 | Patricia                        | Brown                           |
| 4 | Mary                            | Johnson                         |
| 5 | Michael                         | Miller                          |
| 6 | Barbara                         | Wilson                          |
| 7 | David                           | Anderson                        |
| 8 | Robert                          | Jones                           |

12. Find employees who work on all projects controlled by their own department.

```

123 select e.fname, e.lname from employee e
124 where not exists ( select p.pnumber from project p where p.depno = e.depno
125 and p.pnumber not in (select w.pno from works_on w where w.essn = e.ssn));
126
127

```

|   | fname<br>character varying (50) | lname<br>character varying (50) |
|---|---------------------------------|---------------------------------|
| 1 | John                            | Smith                           |
| 2 | Mary                            | Johnson                         |
| 3 | James                           | Williams                        |
| 4 | Robert                          | Jones                           |
| 5 | Linda                           | Davis                           |
| 6 | Barbara                         | Wilson                          |
| 7 | William                         | Moore                           |
| 8 | Elizabeth                       | Taylor                          |
| 9 | David                           | Anderson                        |

13. List the employees who work on every project in the company.

```
128 select e.fname, e.lname from employee e where not exists (select p.pnumber from project p
129 where p.pnumber not in (select w.pno from works_on w where w.essn = e.ssn));
130
```

Data Output Messages Notifications

| fname                  | lname                  |
|------------------------|------------------------|
| character varying (50) | character varying (50) |

14. Find the employees who work on all projects controlled by department number 5.

```
131 select e.fname, e.lname from employee e where not exists (select p.pnumber from project p
132 where p.deptno = 5 and p.pnumber not in (select w.pno from works_on w where w.essn = e.ssn));
133
```

Data Output Messages Notifications

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|   | fname                  | lname                  |
|---|------------------------|------------------------|
|   | character varying (50) | character varying (50) |
| 1 | Elizabeth              | Taylor                 |
| 2 | David                  | Anderson               |